

FIREWALL FORTIGATE Y POLÍTICAS DE SEGURIDAD

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ENTI-UB REDES III

1 CONTENIDO

2	Introducción	2
3	Topología y direccionamiento	2
3.1	Topología	2
3.2	Direccionamiento IP (resumen)	2
4	Dispositivos utilizados	4
5	Configuración del Router (Hardening + RoaS + DHCP)	5
5.1	Hardening aplicado	5
5.2	Enlace de tránsito hacia FortiGate	6
5.3	Router-on-a-Stick (subinterfaces 802.1Q)	7
5.4	DHCP por VLAN	8
6	Configuración del Switch (VLAN + Trunk + Access)	9
6.1	Creación de VLANs	9
6.2	Trunk hacia el router	9
7	Configuración del FortiGate (Interfaces + Routing + NAT)	10
7.1	WAN por DHCP (port1)	10
7.2	Interfaz de tránsito (port2)	10
8	Políticas de seguridad (VLAN10/20/30)	12
9	Bloqueo específico: Instagram en VLAN10	12
10	Logging y monitorización	13
11	Validación final (pruebas por VLAN)	14
11.1	VLAN10 (permitida general + bloqueo Instagram)	14
11.2	VLAN20 (bloqueo total)	14
11.3	VLAN30 (solo WEB/DNS)	15

2 INTRODUCCIÓN

En esta actividad se amplía la infraestructura de la ACT1 integrando un **firewall FortiGate** como perímetro de seguridad. El objetivo es replicar un escenario real corporativo: segmentación interna por VLANs, enrutamiento inter-VLAN mediante Router-on-a-Stick, salida a Internet pasando por el FortiGate y aplicación de políticas de filtrado con logging.

Además, se implementa un **bloqueo específico por dominio (Instagram)** para demostrar filtrado granular, teniendo en cuenta el comportamiento stateful del firewall (sesiones persistentes) y el uso de subdominios.

3 TOPOLOGÍA Y DIRECCIONAMIENTO

3.1 TOPOLOGÍA

Se implementa una salida a Internet centralizada, para ello haremos lo siguiente:

- Red Aula (WAN 192.168.223.0/24) → FortiGate
- Red de tránsito 192.168.224.0/24 → Router Cisco
- Switch → VLANs internas (10.0.10.0/24, 10.0.20.0/24, 10.0.30.0/24)

La red de tránsito evita solapamiento con la red del aula y permite que el FortiGate sea el único equipo expuesto en WAN.

3.2 DIRECCIONAMIENTO IP (RESUMEN)

- **WAN (Aula):** 192.168.223.0/24 (GW 192.168.223.1)
- **FortiGate port1 (WAN):** DHCP (ej. 192.168.223.156)
- **Tránsito FortiGate–Router:** 192.168.224.0/24
 - FortiGate port2: 192.168.224.1/24
 - Router G0/0/0: 192.168.224.2/24
- **VLANs internas:**
 - **VLAN10_FULL:** 10.0.10.0/24 (GW 10.0.10.1)

- **VLAN20_BLOCK:** 10.0.20.0/24 (GW 10.0.20.1)
- **VLAN30_WEB_DNS:** 10.0.30.0/24 (GW 10.0.30.1)

```

/dev/ttyUSB0 - PuTTY
config firewall policy
edit 2
    set name "VLAN30 WEB DNS"
    set uuid 76ee7276-0c17-51f1-718e-875b882b13e5
    set srcintf "port2"
    set dstintf "port1"
    set action accept
    set srcaddr "VLAN30_NET"
    set dstaddr "all"
    set schedule "always"
    set service "HTTP" "HTTPS" "DNS"
    set logtraffic all
    set nat enable
next
edit 10
    set name "BLOCK INSTAGRAM VLAN10"
    set uuid dda77336-0c22-51f1-8191-3b346beb1feb
    set srcintf "port2"
    set dstintf "port1"
    set srcaddr "VLAN10_NET"
    set dstaddr "BLOCK INSTAGRAM"
    set schedule "always"
    set service "HTTP" "HTTPS"
    set logtraffic all
next
edit 1
    set name "VLAN10_FULL"
    set uuid d24a9398-0c1c-51f1-4c95-a6d954fd9b0c
    set srcintf "port2"
    set dstintf "port1"
    set action accept
    set srcaddr "VLAN10_NET"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set logtraffic all
    set nat enable
next
edit 3
    set name "VLAN20_BLOCK"
    set uuid 1989181a-0c1d-51f1-3385-8f13bb347792
    set srcintf "port2"
    set dstintf "port1"
    set srcaddr "VLAN20_NET"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set logtraffic all
next
end
FORTINET-LAB-NICOADRIA #

```

4 DISPOSITIVOS UTILIZADOS

- **Router Cisco C8200L-1N-4T:** gateway por VLAN (subinterfaces 802.1Q), DHCP por VLAN, ruta por defecto hacia FortiGate, hardening.
- **Switch Cisco Catalyst 1300:** VLANs, trunk hacia router, access a clientes.
- **Fortinet FortiGate 121G:** perímetro WAN, NAT, policies por VLAN/servicio, logging, bloqueo FQDN.

```
ceding.  
Please run 'execute disk scan 259'  
Note: The device will reboot and scan the disk during startup. This may take up  
to an hour.  
FORTINET-LAB-NICOADRIA # get system status  
Version: FortiGate-121G v7.0.16,build7538,241028 (GA.M)  
Current Security Level: High  
Physical Switch Security Level: High  
Firmware Signature: certified  
Virus-DB: 1.00000(2018-04-09 18:07)  
Extended DB: 1.00000(2018-04-09 18:07)  
AV AI/ML Model: 0.00000(2001-01-01 00:00)  
IPS-DB: 6.00741(2015-12-01 02:30)  
IPS-ETDB: 0.00000(2001-01-01 00:00)  
APP-DB: 6.00741(2015-12-01 02:30)  
INDUSTRIAL-DB: 6.00741(2015-12-01 02:30)  
IPS Malicious URL Database: 1.00001(2015-01-01 01:01)  
Serial-Number: FG121GTK25005592  
BIOS version: 06000121  
System Part-Number: P28816-20  
Log hard disk: Available  
Hostname: FORTINET-LAB-NICOADRIA  
Private Encryption: Disable  
Operation Mode: NAT  
Current virtual domain: root  
Max number of virtual domains: 10  
Virtual domains status: 1 in NAT mode, 0 in TP mode  
Virtual domain configuration: disable  
FIPS-CC mode: disable  
Current HA mode: standalone  
Branch point: 0667  
Release Version Information: GA  
System time: Tue Feb 17 11:42:18 2026  
Last reboot reason: power cycle  
  
FORTINET-LAB-NICOADRIA #
```


5 CONFIGURACIÓN DEL ROUTER (HARDENING + RoAS + DHCP)

5.1 HARDENING APLICADO

Se aplicaron medidas de seguridad típicas:

- SSH v2
- timeouts en líneas (exec-timeout)
- límite de intentos SSH (authentication-retries)
- control de sesiones (session-limit)
- login local en consola/VTY
- cifrado de contraseñas

```
NICOADRIA(config)#ip ssh vers
NICOADRIA(config)#ip ssh version 2
NICOADRIA(config)#ip ssh
NICOADRIA(config)#ip ssh time
NICOADRIA(config)#ip ssh time-out 60
NICOADRIA(config)#ip ssh
NICOADRIA(config)#ip ssh auth
NICOADRIA(config)#ip ssh authentication-retries 2
NICOADRIA(config)#end
NICOADRIA#wr
Building configuration...
[OK]
NICOADRIA#conf t
Enter configuration commands, one per line. End with CNTL/Z.
NICOADRIA(config)#line
NICOADRIA(config)#line vty 0 4
NICOADRIA(config-line)#login local
NICOADRIA(config-line)#transport input
NICOADRIA(config-line)#transport inpt
NICOADRIA(config-line)#transpor
NICOADRIA(config-line)#transport inp
NICOADRIA(config-line)#transport input ssh telnet
NICOADRIA(config-line)#exe
NICOADRIA(config-line)#exec-tim
NICOADRIA(config-line)#exec-timeout 5 0
NICOADRIA(config-line)#sessi
NICOADRIA(config-line)#session-
NICOADRIA(config-line)#session
NICOADRIA(config-line)#session-li
NICOADRIA(config-line)#session-limit 2
NICOADRIA(config-line)#end
NICOADRIA#conf t
Enter configuration commands, one per line. End with CNTL/Z.
```

```

NICOADRIA(config)#end
NICOADRIA#wr
Building configuration...
[OK]
NICOADRIA#conf t
Enter configuration commands, one per line. End with CNTL/Z.
NICOADRIA(config)#line
NICOADRIA(config)#line vty 0 4
NICOADRIA(config-line)#login local
NICOADRIA(config-line)#transport input
NICOADRIA(config-line)#transport inpt
NICOADRIA(config-line)#transpor
NICOADRIA(config-line)#transport inp
NICOADRIA(config-line)#transport input ssh telnet
NICOADRIA(config-line)#exe
NICOADRIA(config-line)#exec-tim
NICOADRIA(config-line)#exec-timeout 5 0
NICOADRIA(config-line)#sessi
NICOADRIA(config-line)#session-
NICOADRIA(config-line)#session
NICOADRIA(config-line)#session-li
NICOADRIA(config-line)#session-limit 2
NICOADRIA(config-line)#end

```

Nota: en una versión aparece transport input ssh telnet. En un hardening “perfecto” sería solo ssh, pero si tu práctica lo permite, lo dejas así y lo explicas.

5.2 ENLACE DE TRÁNSITO HACIA FORTIGATE

Se configuró una interfaz hacia la red de tránsito:

- Router G0/0/0 = **192.168.224.2/24**
- Ruta por defecto en router apuntando al FortiGate:
ip route 0.0.0.0 0.0.0.0 192.168.224.1

```

interface GigabitEthernet0/0/0
description TRANSITO-FORTINET
ip address 192.168.224.2 255.255.255.0
negotiation auto
!
interface GigabitEthernet0/0/1
description TRUNK-SWITCH
no ip address
negotiation auto
!
interface GigabitEthernet0/0/1.10
encapsulation dot1Q 10
ip address 10.0.10.1 255.255.255.0
!
interface GigabitEthernet0/0/1.20
encapsulation dot1Q 20
ip address 10.0.20.1 255.255.255.0
!
interface GigabitEthernet0/0/1.30
encapsulation dot1Q 30
ip address 10.0.30.1 255.255.255.0
!
interface GigabitEthernet0/0/2
no ip address
negotiation auto
!
interface GigabitEthernet0/0/3
no ip address
negotiation auto
!
ip forward-protocol nd
no ip http server
ip http secure-server
!
ip route 0.0.0.0 0.0.0.0 192.168.224.1

```

5.3 ROUTER-ON-A-STICK (SUBINTERFACES 802.1Q)

Se configuró el enlace al switch como trunk y se crearon subinterfaces:

- **G0/0/1.10** → encapsulation dot1Q 10 → **IP 10.0.10.1/24**
- **G0/0/1.20** → encapsulation dot1Q 20 → **IP 10.0.20.1/24**
- **G0/0/1.30** → encapsulation dot1Q 30 → **IP 10.0.30.1/24**

```
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#
NIC0ADRIA#show ip int
NIC0ADRIA#show ip interface brie
NIC0ADRIA#show ip interface brief
Interface                IP-Address      OK? Method Status Protocol
GigabitEthernet0/0/0    192.168.224.2   YES manual up      up
GigabitEthernet0/0/1    unassigned      YES unset  up      up
GigabitEthernet0/0/10   10.0.10.1      YES manual up      up
GigabitEthernet0/0/20   10.0.20.1      YES manual up      up
GigabitEthernet0/0/30   10.0.30.1      YES manual up      up
GigabitEthernet8/0/2    unassigned      YES unset  down    down
GigabitEthernet8/0/3    unassigned      YES unset  down    down
NIC0ADRIA#
```


5.4 DHCP POR VLAN

Se crearon pools DHCP por VLAN y se excluyeron las .1 (gateway) y rangos reservados.

```
!
!
ip dhcp excluded-address 10.0.10.1
ip dhcp excluded-address 10.0.20.1
ip dhcp excluded-address 10.0.30.1
!
ip dhcp pool VLAN10
network 10.0.10.0 255.255.255.0
default-router 10.0.10.1
dns-server 8.8.8.8
domain-name lab.local
!
ip dhcp pool VLAN20
network 10.0.20.0 255.255.255.0
default-router 10.0.20.1
dns-server 8.8.8.8
domain-name lab.local
!
ip dhcp pool VLAN30
network 10.0.30.0 255.255.255.0
default-router 10.0.30.1
dns-server 8.8.8.8
domain-name lab.local
!
!
login on-success log
!
```

```
- [x] [icon] /dev/ttyUSB1
+ v
[icon] [icon]
are ...
boot-start-marker
boot-end-marker
!
no aaa new-model
!
no ip domain lookup
ip domain name lab.local
!
!
!
ip dhcp excluded-address 10.0.10.1
ip dhcp excluded-address 10.0.20.1
ip dhcp excluded-address 10.0.30.1
!
ip dhcp pool VLAN10
network 10.0.10.0 255.255.255.0
```

6 CONFIGURACIÓN DEL SWITCH (VLAN + TRUNK + ACCESS)

6.1 CREACIÓN DE VLANs

Se crearon VLAN 10, 20 y 30 y se asignaron puertos access a clientes según VLAN.

```
Detected speed: 9600

User Name:
User Name:nicoadria
Password:*****

SWITCH-LAB-NICOADRIA#show vlan
Created by: D-Default, S-Static, G-GVRP, R-Radius Assigned VLAN, V-Voice VLAN

Vlan      Name      Tagged Ports      UnTagged Ports      Created by
-----
1          1          gi1                gi1,gi15-28         D
10         10         gi1                gi2                  S
20         20         gi1                gi3                  S
30         30         gi1                gi4                  S

SWITCH-LAB-NICOADRIA#
```

6.2 TRUNK HACIA EL ROUTER

El puerto hacia el router se forzó en modo trunk y se permitieron VLANs 10,20,30.

```
SWITCH-LAB-NICOADRIA#
SWITCH-LAB-NICOADRIA#
SWITCH-LAB-NICOADRIA#
SWITCH-LAB-NICOADRIA#
SWITCH-LAB-NICOADRIA#show interfaces switchport gi1
S-VLAN Ethernet Type: 802.1q
VLAN Mapping Tunnel L2 protocols Global CoS: 5
Name: gi1
Switchport: enable
Administrative Mode: trunk
Operational Mode: up
Access Mode VLAN: 1
Access Multicast TV VLAN: none
Trunking Native Mode VLAN: 1
Trunking VLANs: 1,10,20,30
                2-9,11-19,21-29,31-4094 (Inactive)
General PVID: 1
General VLANs: none
General Egress Tagged VLANs: none
General Forbidden VLANs: none
General Ingress Filtering: enabled
General Acceptable Frame Type: all
General GVRP status: disabled
Customer Mode VLAN: none
Customer Multicast TV VLANs: none
Private-vlan promiscuous-association primary VLAN: none
Private-vlan promiscuous-association Secondary VLANs: none
Private-vlan host-association primary VLAN: none
Private-vlan host-association Secondary VLAN: none

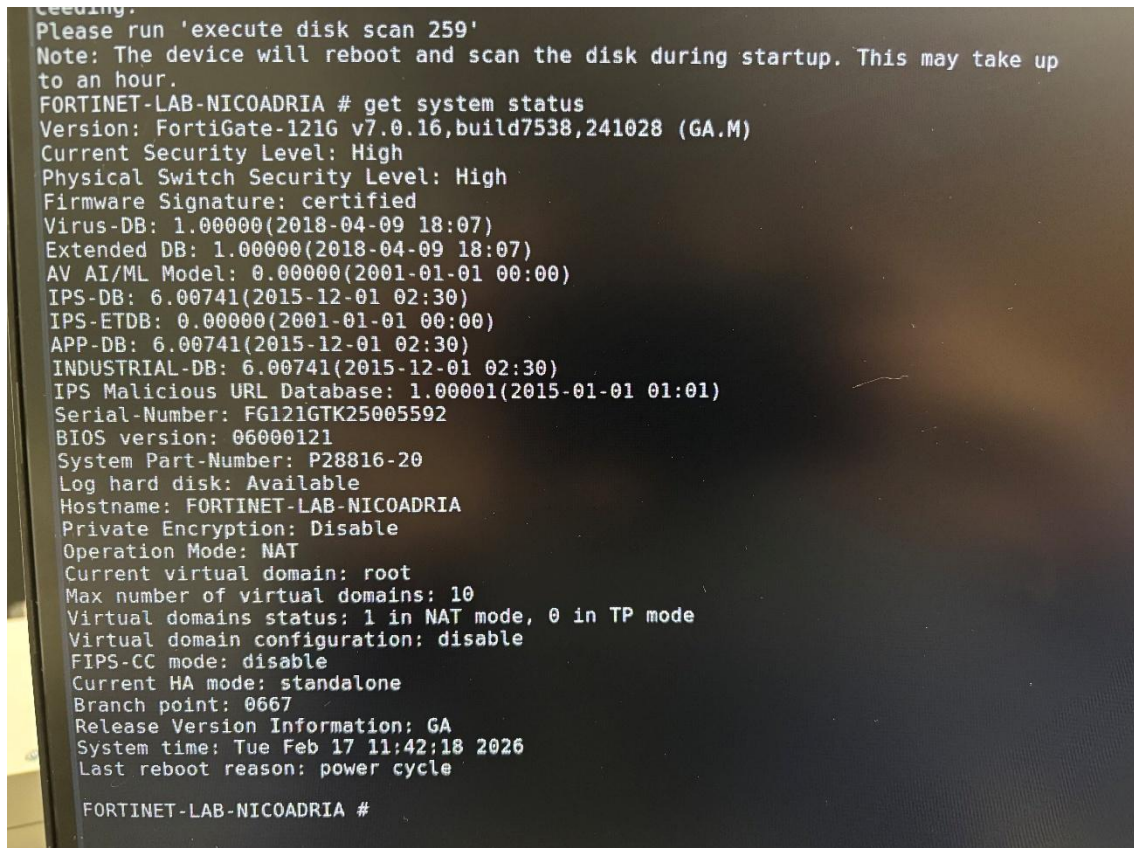
VLAN Mapping Tunnel - no resources
VLAN Mapping One-To-One - no resources

Classification rules:
Classification type Group ID VLAN ID
-----
SWITCH-LAB-NICOADRIA#
```

7 CONFIGURACIÓN DEL FORTIGATE (INTERFACES + ROUTING + NAT)

7.1 WAN POR DHCP (PORT1)

El FortiGate se conectó a la red del aula (WAN) configurando port1 en modo DHCP.



```
ceding.  
Please run 'execute disk scan 259'  
Note: The device will reboot and scan the disk during startup. This may take up  
to an hour.  
FORTINET-LAB-NICOADRIA # get system status  
Version: FortiGate-121G v7.0.16,build7538,241028 (GA.M)  
Current Security Level: High  
Physical Switch Security Level: High  
Firmware Signature: certified  
Virus-DB: 1.00000(2018-04-09 18:07)  
Extended DB: 1.00000(2018-04-09 18:07)  
AV AI/ML Model: 0.00000(2001-01-01 00:00)  
IPS-DB: 6.00741(2015-12-01 02:30)  
IPS-ETDB: 0.00000(2001-01-01 00:00)  
APP-DB: 6.00741(2015-12-01 02:30)  
INDUSTRIAL-DB: 6.00741(2015-12-01 02:30)  
IPS Malicious URL Database: 1.00001(2015-01-01 01:01)  
Serial-Number: FG121GTK25005592  
BIOS version: 06000121  
System Part-Number: P28816-20  
Log hard disk: Available  
Hostname: FORTINET-LAB-NICOADRIA  
Private Encryption: Disable  
Operation Mode: NAT  
Current virtual domain: root  
Max number of virtual domains: 10  
Virtual domains status: 1 in NAT mode, 0 in TP mode  
Virtual domain configuration: disable  
FIPS-CC mode: disable  
Current HA mode: standalone  
Branch point: 0667  
Release Version Information: GA  
System time: Tue Feb 17 11:42:18 2026  
Last reboot reason: power cycle  
  
FORTINET-LAB-NICOADRIA #
```

7.2 INTERFAZ DE TRÁNSITO (PORT2)

Se configuró port2 como red de tránsito hacia router:

- port2: **192.168.224.1/24**

6.3 Rutas estáticas necesarias

Rutas típicas:

- Default route hacia WAN (normalmente por DHCP)
- Ruta de retorno hacia redes internas:
10.0.0.0/8 vía 192.168.224.2 por port2


```

/dev/ttyUSB0 - PuTTY
config firewall policy
edit 2
    set name "VLAN30 WEB DNS"
    set uuid 76ee7276-0c17-51f1-718e-875b882b13e5
    set srcintf "port2"
    set dstintf "port1"
    set action accept
    set srcaddr "VLAN30_NET"
    set dstaddr "all"
    set schedule "always"
    set service "HTTP" "HTTPS" "DNS"
    set logtraffic all
    set nat enable
next
edit 10
    set name "BLOCK INSTAGRAM_VLAN10"
    set uuid dda77336-0c22-51f1-8191-3b346beb1feb
    set srcintf "port2"
    set dstintf "port1"
    set srcaddr "VLAN10_NET"
    set dstaddr "BLOCK INSTAGRAM"
    set schedule "always"
    set service "HTTP" "HTTPS"
    set logtraffic all
next
edit 1
    set name "VLAN10 FULL"
    set uuid d24a9398-0c1c-51f1-4c95-a6d954fd9b0c
    set srcintf "port2"
    set dstintf "port1"
    set action accept
    set srcaddr "VLAN10_NET"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set logtraffic all
    set nat enable
next
edit 3
    set name "VLAN20_BLOCK"
    set uuid 1989181a-0c1d-51f1-3385-8f13bb347792
    set srcintf "port2"
    set dstintf "port1"
    set srcaddr "VLAN20_NET"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set logtraffic all
next
end
FORTINET-LAB-NICOADRIA #
6:32 PM 2/17/26
```


8 POLÍTICAS DE SEGURIDAD (VLAN10/20/30)

Se definieron políticas por VLAN:

- **VLAN10_FULL**: allow all con NAT
- **VLAN20_BLOCK**: deny all
- **VLAN30_WEB_DNS**: allow solo HTTP/HTTPS/DNS con NAT

Importante: En FortiGate el **orden** manda (top-down). Un allow general por encima se come el deny específico.

```
set end-ip 10.212.134.210
next
edit "FCTEMS_ALL_FORTICLOUD_SERVERS"
set uuid eb970ea8-0b4f-51f1-762b-c60f7a4617da
set type dynamic
set sub-type ems-tag
next
edit "VLAN10_NET"
set uuid ff765d80-0c16-51f1-7eae-074e48af5dbb
set subnet 10.0.10.0 255.255.255.0
next
edit "VLAN20_NET"
set uuid 099ac954-0c17-51f1-d989-1324e2c3e70f
set subnet 10.0.20.0 255.255.255.0
next
edit "VLAN30_NET"
set uuid 27869672-0c18-51f1-ff25-6547c0181915
set subnet 10.0.30.0 255.255.255.0
next
end
FORTINET-LAB-NICOADRIA #
```

9 BLOQUEO ESPECÍFICO: INSTAGRAM EN VLAN10

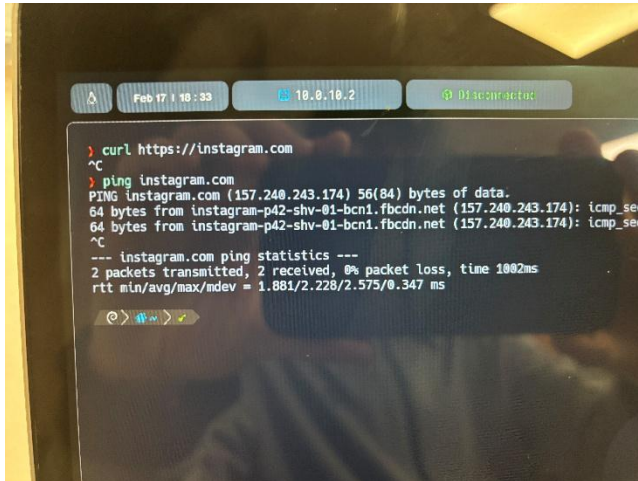
Para bloquear Instagram se creó un objeto tipo FQDN con wildcard:

- *.instagram.com

Luego se creó una policy **deny** para VLAN10 contra ese destino, y se colocó **por encima** de la policy allow general.

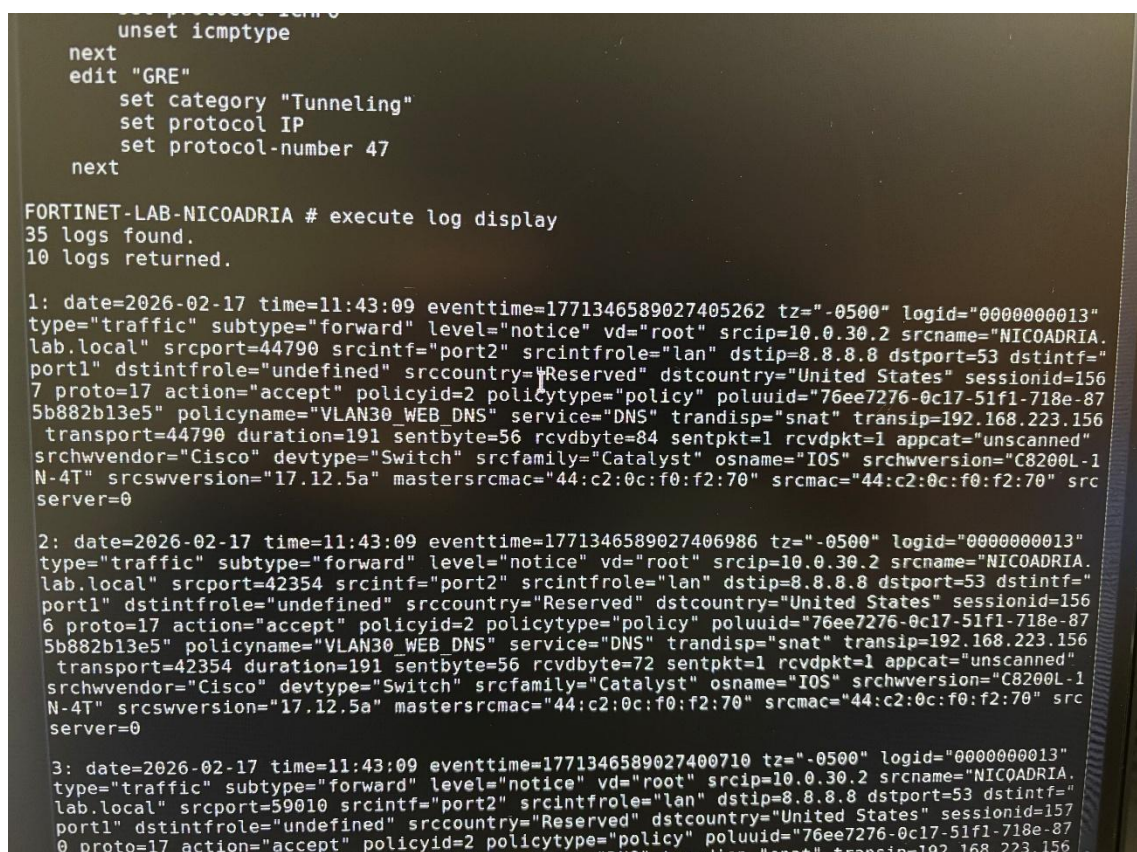
Finalmente se limpió la tabla de sesiones:

- diagnose sys session clear



10 LOGGING Y MONITORIZACIÓN

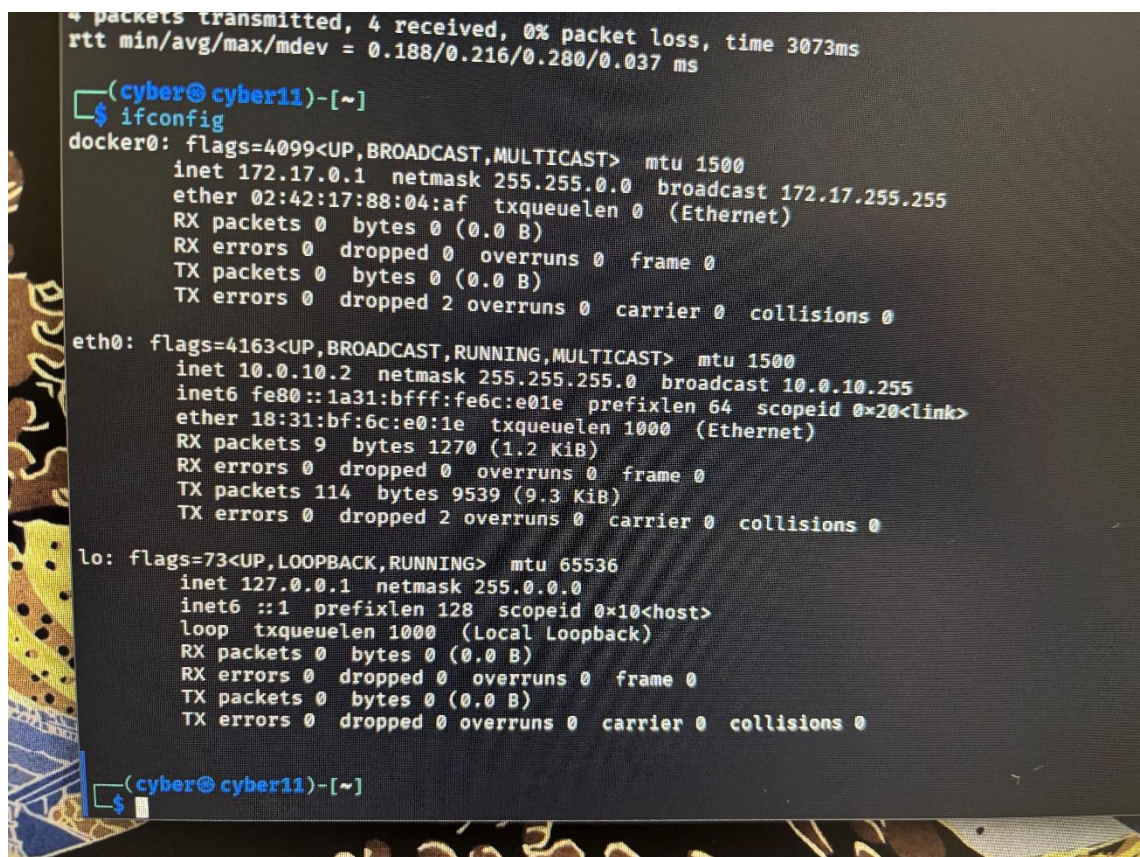
Se habilitó logging en políticas (logtraffic all) y se revisaron logs para comprobar acciones **accept/deny**.



11 VALIDACIÓN FINAL (PRUEBAS POR VLAN)

11.1 VLAN10 (PERMITIDA GENERAL + BLOQUEO INSTAGRAM)

- ping 8.8.8.8 ✓
- DNS ✓
- HTTP/HTTPS ✓
- Instagram ✗



```
4 packets transmitted, 4 received, 0% packet loss, time 3073ms
rtt min/avg/max/mdev = 0.188/0.216/0.280/0.037 ms

(cyber@cyber11)-[~]
$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:42:17:88:04:af txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.10.2 netmask 255.255.255.0 broadcast 10.0.10.255
    inet6 fe80::1a31:bfff:fe6c:e01e prefixlen 64 scopeid 0<20<link>
    ether 18:31:bf:6c:e0:1e txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 1270 (1.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 114 bytes 9539 (9.3 KiB)
    TX errors 0 dropped 2 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(cyber@cyber11)-[~]
```

11.2 VLAN20 (BLOQUEO TOTAL)

- ping ✗
- DNS ✗
- HTTP/HTTPS ✗


```
~: zsh
New Tab Split View
(cyber@cyber11)-[~]
$ ping 10.0.10.1
PING 10.0.10.1 (10.0.10.1) 56(84) bytes of data.
64 bytes from 10.0.10.1: icmp_seq=1 ttl=255 time=0.280 ms
64 bytes from 10.0.10.1: icmp_seq=2 ttl=255 time=0.188 ms
64 bytes from 10.0.10.1: icmp_seq=3 ttl=255 time=0.191 ms
64 bytes from 10.0.10.1: icmp_seq=4 ttl=255 time=0.205 ms
^C
— 10.0.10.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3073ms
rtt min/avg/max/mdev = 0.188/0.216/0.280/0.037 ms
(cyber@cyber11)-[~]
```

11.3 VLAN30 (SOLO WEB/DNS)

- DNS ✓
- HTTP/HTTPS ✓
- ping ✗ (si lo bloqueaste por policy/servicio)

```
-4 use IPv4
-b allow pinging broadcast
-R record route
-T <timestamp> define timestamp, can be one of <tsonly/tsandaddr/tsprespec>

IPv6 options:
-6 use IPv6
-F <flowlabel> define flow label, default is random
-N <nodeinfo opt> use IPv6 node info query, try <help> as argument

For more details see ping(8).

(cyber@cyber11)-[~]
$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1 10.0.10.1 (10.0.10.1) 0.245 ms 0.172 ms 0.202 ms
 2 192.168.224.1 (192.168.224.1) 0.169 ms 0.185 ms 0.184 ms
 3 192.168.223.1 (192.168.223.1) 0.376 ms 0.344 ms 0.313 ms
 4 172.26.3.1 (172.26.3.1) 0.788 ms 0.790 ms 0.758 ms
 5 254.pool85-50-130.static.orange.es (85.50.130.254) 3.096 ms 2.490 ms 3.067 ms
 6 10.255.119.113 (10.255.119.113) 3.033 ms 3.345 ms 4.315 ms
 7 10.34.217.45 (10.34.217.45) 2.813 ms 2.774 ms 10.34.217.49 (10.34.217.49) 2.790 ms
 8 10.34.200.21 (10.34.200.21) 2.694 ms 2.140 ms 2.024 ms
 9 10.34.65.173 (10.34.65.173) 1.975 ms 1.933 ms 10.34.65.181 (10.34.65.181) 1.961 ms
10 * 81.52.188.113 (81.52.188.113) 2.250 ms 3.194 ms
11 193.251.129.205 (193.251.129.205) 11.366 ms 11.335 ms 10.200 ms
12 72.14.204.60 (72.14.204.60) 10.129 ms 72.14.209.130 (72.14.209.130) 10.165 ms 74.125.48.64 (74.125.48.64) 10.176 ms
13 142.251.249.207 (142.251.249.207) 10.418 ms 192.178.110.155 (192.178.110.155) 12.498 ms 108.176.14.10 (108.176.14.10) 10.176 ms
14 142.251.49.55 (142.251.49.55) 21.937 ms 142.250.46.165 (142.250.46.165) 12.781 ms 142.251.51.14 (142.251.51.14) 10.176 ms
15 dns.google (8.8.8.8) 10.757 ms 10.596 ms 10.771 ms

(cyber@cyber11)-[~]
$
```